Which Comparison Counts? Benchmarking and Belief Updating

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Abstract

Field experiments have mixed findings on how and to what extent information changes voting behavior. Many factors may explain these mixed results, but one that remains under-explored is how benchmarking, or the presentation of performance information relative to other governments, can change interpretation. In a pre-registered survey fielded in Mexico with both descriptive and experimental components, I examine how benchmarking changes voting behavior. I find that respondents prefer benchmark municipalities that are geographically proximate, highly populated, and governed by the same party coalition as the respondent's municipality. When randomly assigned to treatment conditions that vary whether partisan benchmarks are shown to respondents, I find that only same-coalition comparisons change respondents' intentions to vote for an incumbent. This effect is driven in part by respondents updating negatively about incumbent politicians relative to opposition party coalitions.

Word count: 6,881

Introduction

Understanding the way voters interpret and process information is essential for both theoretical and empirical research that examines electoral accountability and government responsiveness. A large body of research in the democracies of developing countries examines whether information about politicians' performance affects electoral outcomes, but findings are extremely mixed (Ferraz and Finan 2008; Chong et al. 2015; Boas and Hidalgo 2011). There are many potential reasons for these mixed findings that have been explored, but less attention has been paid to how benchmarking of information itself may affect voting behavior.Benchmarks are very commonly used to make information interpretable in informational campaigns (Grossman and Slough 2022; Keefer and Khemani 2005), but limited theoretical or empirical study has been done to understand how benchmark construction matters.

The Metaketa I project (Dunning et al. 2019), a coordinated set of pre-registered information experiments, illustrates this lack of attention to benchmarking. Table 1 summarizes the information and the benchmark used in the four experiments that shared performance information about politicians. Even within this deliberately harmonized design, the choice of benchmark varies considerably across studies. Only the pre-analysis plan for the experiment in Mexico explicitly considers how providing a benchmark differs from providing non-benchmarked information, and in all four studies the reason for choosing the specified benchmark is not made explicit. This is less a shortcoming of any individual study than a reflection of how little attention the concept has received.

To address the lack of attention on benchmarks, I present results from a pre-registered survey in Mexico to understand how voters in developing countries use information when deciding which candidate or political party to vote for. The survey has two components. First, I design a comparison municipality choice exercise to examine which municipalities respondents prefer as benchmarks. In the survey, those their preferred benchmark municipalities from a candidate menu of 15 municipalities constructed to show municipalities that

Table 1: Information and benchmarks used in the Metaketa I studies (Dunning et al. 2019)

Study	Information provided	Benchmark
Benin	Legislative performance	Other legislators in the region and country
Mexico	Municipal unauthorized or mis-allocated spending	Municipalities in the same state governed by a different party
Uganda II	Budget irregularities	Other districts within country
Brazil	Municipal accounts rejected by independent audit	Other within-state municipalities

are likely to be relevant to respondents. I find that respondents prefer benchmark municipalities that are geographically proximate, from the same state, highly populated, and governed by the same coalition as the respondent's municipality. The preference for same-coalition municipalities appears to is in part a result of respondents' actual knowledge of which parties govern nearby municipalities, rather than unmeasured dimensions of similarity between municipalities.

Second, I run an experiment in which respondents are randomly assigned to receive information about non-violent crime rates in treatment conditions that vary whether municipalities shown as benchmarks are labeled as from the same coalition as their municipal government, from a different coalition, or unlabeled. I use crime because it is highly relevant to Mexican voters and because municipal presidents largely control police forces. I find that only same-coalition comparisons change respondents' intentions to vote for an incumbent. The observed effect is robust to a variety of specifications. I show that the effect is partially explained by changes in beliefs about incumbents' crime handling.

These results point to two major implications for research on electoral accountability with informational treatments. First, varying the benchmark presented to voters can change how voters respond to the information, which may explain some of the mixed results in the literature. Second, same-coalition comparisons are theoretically well suited to filtering noise out of performance information, and I find that voters treat them as more relevant than opposite-coalition comparisons. Despite this, they are rarely used in informational campaigns and deserve stronger consideration in future research.

Background: benchmarking in political science

Scholars of economic voting have examined how economic performance relative to differing benchmarks affects voter behavior. These studies generally find that accounting for how voters interpret economic information is crucial for understanding the relationship between economic outcomes and voting behavior. Past work has found that interpretation of economic policies and outcomes can be altered by comparisons between sub-national governments (Besley and Case 1995), comparisons from local to national performance (Cohen 2020), and performance relative to expectations (Epp and Wlezien 2026). Observational studies comparing whether economic performance relative to international benchmarks can explain domestic vote choice have mixed results (Kayser and Peress 2012; Aytaç 2018; Arel-Bundock, Blais, and Dassonneville 2021), but experimental studies that explicitly show cross-national comparisons to respondents have shown that these benchmarks strongly influence voters' perceptions of performance (Huang 2015; Hansen, Olsen, and Bech 2015).

There is reason to believe that the importance of benchmarking in the literature cited above would also hold in developing democracies across different performance areas. Voters across many developing democracies respond to information about incumbent performance when it is relevant to them (Gottlieb 2016; Adida et al. 2020; Banerjee et al. 2024). Specifically, crime is a salient valence issue on which voters evaluate incumbents (Singer 2011), and in Latin American contexts voters have been shown to hold politicians accountable for insecurity when they can attribute responsibility for it (Ley 2017). Additionally, voting is not driven solely by clientelism: voters in developing democracies do not base their preferences solely on clientelistic favors (Cruz, Keefer, and Labonne 2021; Cruz et al. 2024). Across these performance areas, the effect of information hinges on whether voters can interpret it. This makes how that information is benchmarked a central question.

The most direct precedent for studying benchmarks in a developing democracy is Arias et al. (2024). They find that information about performance relative to other municipalities does not influence voters' beliefs about their own municipality's performance in Mexico. However, while they compare benchmarked to non-benchmarked information, they do not test whether the chosen benchmark was relevant to voters. This paper takes up that question by testing the relevance of different comparisons (e.g., same-coalition vs. opposite-coalition) and directly observing which comparisons voters find most relevant. The design directly addresses the lack of consensus on how to construct benchmarks, despite their widespread use (Grossman and Slough 2022; Keefer and Khemani 2005).

A model of benchmarking performance information

To understand how benchmarks may help voters interpret noisy performance information, I present a voter decision theoretic model that builds on the framework of Besley and Case (1992). The central problem facing voters is that most observable performance outcomes, like crime rates, are driven by both the incumbent's effort and factors outside any politician's control (e.g., economic shocks or regional crime trends that affect entire areas simultaneously). Because these common factors are unobserved, a voter who sees only their municipality's crime rate cannot tell whether a high crime rate reflects poor governance or "bad luck" (factors out of politicians' control that increase crime).

To illustrate this logic formally, consider the simple case where the crime rate r_h in the voter's home municipality h is a factor of government competence, z_h, and an exogenous shock, θ, that is common to municipalities in the same region.

$$r_h = \theta + z_h$$

The exogenous shock θ can take a value of 0 (no shock), 1 (small shock), or 2 (large shock). Competence z_h takes on a value of 0 (competent) or 1 (incompetent).

There are four possible realizations of crime rates: $r_h \in \{0, 1, 2, 3\}$. Of these four possible crime rates, only the extremes reveal competence information. The crime rate $r_h = 0$ can only be achieved by a competent government and the crime rate $r_h = 3$ can only be achieved by an incompetent government. For the crime rates in the ambiguous set $\mathcal{A} = \{1, 2\}$, the voter must rely on their prior beliefs about the prevalence of incompetence and the likelihood of a small shock.

The informational value of the crime rate is enhanced when the voter observes their home crime rate against a benchmark municipality's crime rate r_b along with their own municipality's crime rate r_h, where b experiences the same shock θ as the home municipality. In this case, the middling crime cases can be fully revealing if the municipal governments have different levels of competence. Consider the case where the politician in the home municipality is incompetent ($z_h = 1$) and the politician in the benchmark municipality is competent ($z_b = 0$). For any value of θ, the difference between municipalities simplifies to reveal just the differences in competence. Formally, this can be shown as $r_h - r_b = z_h - z_b = 1$, which fully reveals the home incumbent's incompetence. Thus, adding a benchmark is weakly more informative for all values of r_h.

Adding policy preferences. The baseline model above assumes that voters do not have preferences for things other than the level of crime. I will now consider cases where policy differences across candidates play a role in voter behavior. Suppose there are two policies x: one policy has no tolerance for crime ($x = 0$), and the other policy tolerates some crime ($x = 1$) in order to dedicate resources to other policy goals. Crime in municipality $m \in \{h, b\}$ is now

$$r_m = \theta + x_m + z_m$$

where x_m is the policy of the politician governing m. Candidates' x is unknown to the voter.

In this case, the possible values for the crime rate are now $r_h \in \{0, 1, 2, 3, 4\}$, and the ambiguous set expands to $\mathcal{A} = \{1, 2, 3\}$. Again, with no benchmark only the extremes are fully revealing. $r_h = 0$ means the politician is a competent, zero tolerance candidate and $r_h = 4$ means the politician is an incompetent, tolerant candidate. Any other case is ambiguous. For example, in the case of $r_h = 1$, this outcome could be observed under a competent zero tolerance candidate, an incompetent zero tolerance candidate, or a competent tolerant candidate. For $r_h = 2$, any combination of policy preference and competence is possible. Introducing policy differences introduces more ambiguity to the voter's inference problem for every candidate type, making inferences drawn from observing the crime rate even more dependent on voter priors about politicians.

A simple comparison of crime rates across municipalities no longer isolates competence, as it now reveals differences in x and z instead of isolating z. Formally, $r_h - r_b = (x_h - x_b) + (z_h - z_b)$. The comparison still improves on plain information, because it can reveal both policy and competence in some ambiguous cases. Consider the case where $(x_h, z_h) = (1, 1)$ and $(x_b, z_b) = (0, 0)$. Here, $r_h - r_b = 2$, which fully reveals the policy preference ($x_h = 1$) and competency ($z_h = 1$) of the incumbent. Writing $d = r_h - r_b \in \{-2, -1, 0, 1, 2\}$ for the observed difference, it is the extremes $d = \pm 2$ that are revealing in this way. The intermediate values are not: $d = \pm 1$ tells the voter that the two municipalities differ, but not whether the difference is one of policy or of competence, and $d = 0$ is consistent both with candidates who match on both dimensions and with candidates whose policy and competence differences offset each other.

Partisan comparisons. I will now consider that candidates come from one of two political parties, A and B, with policy positions x_A and x_B. Let $P(m) \in \{A, B\}$ denote the party governing municipality m, so that a candidate's policy is fixed by their party, $x_m = x_{P(m)}$. However, the parties need not differ in their policies, and individual candidates can vary on competence z_m within party. In other words, two candidates from the same party can differ in competence but must have the same policy preference, while two candidates from different parties can (but need not) differ on both policy and competence. Assume that the voter knows which party governs their municipality, but does not know which party governs a comparison municipality unless informed.

The addition of political parties does not change the informational context for the non-benchmarked case nor for the non-partisan benchmark, because the voter will not know which party governs the comparison municipality. If the benchmark includes information about the comparison municipality's party, however, more information can be gleaned from the benchmark. If the benchmark municipality is revealed to be governed by the same party as the home municipality ($P(h) = P(b)$), then $x_h = x_b$ and $d = z_h - z_b$. The same-party comparison therefore returns the voter to the baseline model, in which the benchmark isolates competence when differences are observed.

For a cross-party comparison ($P(h) \neq P(b)$), the benchmark reveals a combination of policy and competence differences, $d = (x_h - x_b) + (z_h - z_b)$, just as in the case without party information. What the party label adds is the knowledge that the two candidates are drawn from different parties, and therefore that $x_h - x_b$ may be nonzero. By itself this does not resolve the confound: $d = \pm 2$ requires both components to take the same extreme value and so reveals policy and competence together, $d = \pm 1$ is consistent either with a policy difference and equal competence or with equal policy and a competence difference, and $d = 0$ is consistent both with candidates who match on both dimensions and with candidates whose differences offset.

The label becomes informative about competence when the voter holds a *directional* prior about how the two parties differ. Put differently, the opposition comparison can increase informativeness when the voter believes not only that the parties differ on crime policy but which of the two is the stricter one. Suppose the voter is confident that $x_h - x_b = \delta$ for a known $\delta \in \{-1, 1\}$. Then $d = (x_h - x_b) + (z_h - z_b) = z_h - z_b - \delta$. Because the competence difference must lie in $\{-1, 0, 1\}$, only three values of d can be observed, two of which pin down competence exactly. Take $\delta = -1$, the case in which the home municipality is governed by the stricter party. Observing $d = 0$ implies $z_h - z_b = 1$, meaning $(z_h, z_b) = (1, 0)$. In other words, the home municipality matched the benchmark's crime rate despite the advantage of a stricter policy, which is possible only if its incumbent is incompetent and the benchmark's is not. Only $d = \delta$, the difference that exactly matches the policy gap, leaves competence unresolved, and even there the voter learns that the two incumbents are equally competent, though not at what level.

Without a directional prior on policy differences, the cross-party benchmark is no more informative about competence than the unlabeled one. A voter who does not know whether the parties differ on crime policy, or who does not know in which direction, cannot separate $x_h - x_b$ from $z_h - z_b$ in any realization of d. Such a voter may still learn about policy rather than competence, but only noisily, since competence is equally likely in both parties and a single comparison cannot distinguish a party-level policy gap from an idiosyncratic competence gap.

These two cases assign the partisan arms distinct roles. The same-party comparison delivers competence information alone, so its effect should operate entirely through the competence channel and should not depend on what the voter believes about the parties' crime policies. The cross-party comparison is the only benchmark that carries information about policy, and it identifies competence in ambiguous cases only for voters holding a confident directional prior about which party is tougher on crime. The model therefore predicts the same-party comparison to cleanly provide information about competence. The cross-party comparison is predicted to show information about competence depending on voters' beliefs about how the parties differ. Furthermore, it is only in the cross-party case that a voter who cares primarily about policy alignment has anything to learn.

Application to the survey. The model depends on the voter accepting the benchmark as an appropriate standard of comparison. If the voter does not believe that the benchmark municipality plausibly faced conditions similar to their own, there is no clean common shock to difference out. A benchmark the voter does find relevant, by contrast, provides the comparison needed to differentiate exogenous shocks from incumbent performance. Understanding which comparisons voters treat as relevant is therefore an important step necessary for making predictions about how voters react to information. The first wave of the survey measures this directly by eliciting the benchmarks respondents would choose for themselves.

The experimental design allows me to test whether comparisons help voters attribute outcomes to incumbent performance rather than to factors outside the incumbent's control. It also tests for whether partisan benchmarks differ from non-partisan benchmarks. Specifically, the difference between same- and opposition-party comparisons can clarify whether voters are focused on re-electing competent parties or learning about parties' platforms.

It is a real question of which of these two approaches applies to Mexican voters. On the one hand, political parties and coalitions in Mexican municipalities are quite distinct. There are three main coalitions: MORENA/PVEM/PT, PAN/PRI/PRD, and Movimiento Ciudadano. Of all 2469 municipal governments in 2026, only 5 rural municipalities were governed by parties from two of the three major coalitions (see Appendix section 4). Thus, voters may already have strong priors about how parties differ on crime and therefore only be interested in implied differences in competence. On the other hand, MORENA's embracing anti-crime policies under Sheinbaum may mean that voters are genuinely unsure which party in 2026 is tougher on crime.

Survey Design

This project involves conducting a survey with two parts. First, a descriptive analysis of the municipalities that are chosen by survey respondents as comparisons for their own municipality. Second, an experiment that varies whether partisan information is shown in benchmarked informational treatments to understand how informational benchmarks affect perceptions of incumbent politicians. Both designs were pre-registered on OSF.

The survey is fielded in two waves. Wave 1 collects pre-treatment beliefs and elicits respondents' preferences for comparison municipalities. Wave 2 recontacts the same respondents approximately one week after completion of Wave 1 and assigns respondents to treatment, then collects post-treatment beliefs.

The two components address complementary halves of the argument. The descriptive choice-based analysis characterizes which benchmarks respondents treat as relevant. The experimental component then manipulates benchmark presentation, varying whether a comparison is shown at all and whether the partisan identity of the comparison municipality's governing coalition is made visible, in order to test how these features shape belief updating. The specific questions the experiment is designed to answer are stated in the Experimental Design section below.

Sample Composition

The study population is eligible voters in the 28 states holding municipal elections in 2027 for which the partisan treatments are well defined.[1] Within this frame, wave 1 respondents were recruited against marginal quotas on sex, age, socioeconomic level (SEL), and state. These quotas were enforced at the start of fielding but relaxed as collection proceeded in order to reach the target sample size, so the achieved wave 1 sample of 3,028 respondents differs slightly from population demographics. Wave 2 was not quota-sampled: it recontacted respondents who completed wave 1, and the 1,927 respondents in the wave 2 analysis sample are simply those who returned, identified the same home municipality in both waves, and passed the attention check. Wave 2's composition therefore reflects wave 1's composition together with differential attrition.[2]

1. Four states are excluded: Durango and Veracruz hold no 2027 municipal elections, Mexico City is excluded because its borough presidents lack municipal police authority, and Oaxaca is excluded because a majority of its municipalities govern under *usos y costumbres*, which makes partisan treatments inapplicable.

2. Sex and state benchmarks are computed from INEGI 2020 census counts for the 28 states in the frame; age benchmarks are the INEGI 2020 distribution of the 18+ population in those states; SEL benchmarks are AMAI NSE shares from ENIGH 2022 (*Estimaciones NSE 2022* 2022). Because the SEL quotas set for wave 1 were themselves not proportional to the population distribution, the comparisons below are drawn against population shares rather than against the field quotas.

On sex and age the departures are modest. Women are overrepresented by 2.5 percentage points in wave 1 (53.6% versus 51.0%) and 2.8 points in wave 2. Age is close to the population distribution in wave 1, where no bracket deviates by more than 2 points, while wave 2 skews somewhat older: the 18–24 bracket falls 2.3 points below its population share and the 45–54 and 55–64 brackets each run roughly 2 points above.

The departures on socioeconomic level and geography are much larger, and they run in the same direction. Both samples tilt toward wealthier, more urban, and more central respondents in both waves. This has no bearing on the internal validity of the experiment, since treatment is randomized within the realized sample, but it does show that the poorest and most rural segments of the Mexican electorate aren't particularly well represented among respondents.

Benchmark Municipality Selection Analysis

In wave 1 of the survey, I seek to identify which comparison municipalities respondents view as most relevant. To do this, I design a candidate-municipality choice exercise. Respondents were given this instruction: "Municipal governments are often evaluated by comparing them to others. Which of the following municipalities would you compare [*name of respondent's municipality*] to?" Below this instruction respondents were shown a list of 15 candidate municipalities before treatment and asked to select any they would like to use as a comparison to their own municipality. Respondents were required to select at least one municipality, and could choose all 15. The 15 candidates are drawn from three pools: 5 randomly sampled from 10 geographically near municipalities (with a population of at least 22,000), 5 from the 20 most populous municipalities nationally, and 5 drawn at random from the full universe. Duplicates across pools are dropped and replaced by resampling. Figure 1 shows an example of how this selection tool looks to respondents. This design gives respondents a manageable list of candidate municipalities that does not systematically privilege any single comparison dimension (geographic proximity, city size, or randomness).

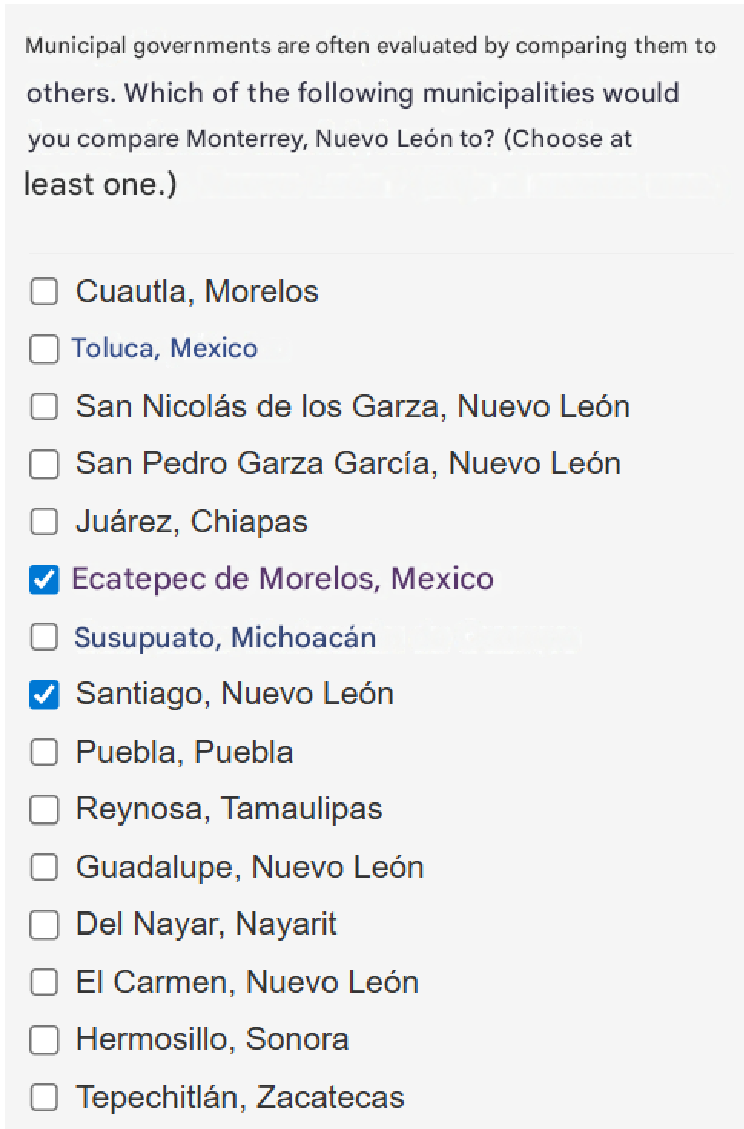
To analyze which municipalities respondents select, I estimate a Bayesian hierarchical logistic regression with random effects, or a Generalized Linear Mixed Model (GLMM). Each of the 15 municipalities shown to respondent r is modeled as an independent Bernoulli trial:

$$\Pr(Y_{ri} = 1) = \text{logit}^{-1}(\alpha_r + \mathbf{X}_{ri}\boldsymbol{\beta}) \tag{1}$$

where $Y_{ri} = 1$ if respondent r selects municipality i and $\mathbf{X}_{ri}$ is a vector of municipality-level covariates (log distance from home municipality, log population ratio of comparison relative to home municipality, same-state indicator, same-coalition indicator, an indicator for whether the candidate municipality's governing coalition matches the respondent's pre-treatment vote intention coalition, and the candidate municipality's governing coalition). Following McCartan, Brown, and Imai (2024), $\alpha_r \sim \mathcal{N}(0, \sigma_\alpha^2)$ is a respondent-level random intercept capturing heterogeneity in total number of municipalities selected. A pool fixed effect (nearest / largest / random) is included to partial out mechanical differences in selection rates across pools.

Prior distributions for the coefficients are formed indirectly by taking a QR decomposition of the centered covariate matrix (Goodall 1993). Coefficients on the QR-decomposed centered covariates are assigned weakly informative $t_2(0, 2.5)$ priors, which implies priors for the actual coefficients that are adapted to the scale and correlation structure of the covariates. Because the 15-municipality menu is researcher-constructed, estimated coefficients are conditional on the offered menu and should be interpreted accordingly.

Results from the 3,028 respondents from wave 1 are shown in Table 2 and Figure 2. The figure plots posterior mean percentage point changes in selection probability relative to a 50-50 baseline. Several patterns stand out. First, spatial variables tend to matter, as respondents are less likely to select municipalities that are farther away from their municipality and more likely to select those from the same state. Second, respondents are more likely to select municipalities that are larger than their home municipality. This effect is


Municipal governments are often evaluated by comparing them to others. Which of the following municipalities would you compare Monterrey, Nuevo León to? (Choose at least one.)

- [ ] Cuautla, Morelos
- [ ] Toluca, Mexico
- [ ] San Nicolás de los Garza, Nuevo León
- [ ] San Pedro Garza García, Nuevo León
- [ ] Juárez, Chiapas
- [x] Ecatepec de Morelos, Mexico
- [ ] Susupuato, Michoacán
- [x] Santiago, Nuevo León
- [ ] Puebla, Puebla
- [ ] Reynosa, Tamaulipas
- [ ] Guadalupe, Nuevo León
- [ ] Del Nayar, Nayarit
- [ ] El Carmen, Nuevo León
- [ ] Hermosillo, Sonora
- [ ] Tepechitlán, Zacatecas

Figure 1: An example of the comparison municipality selection question shown to respondents

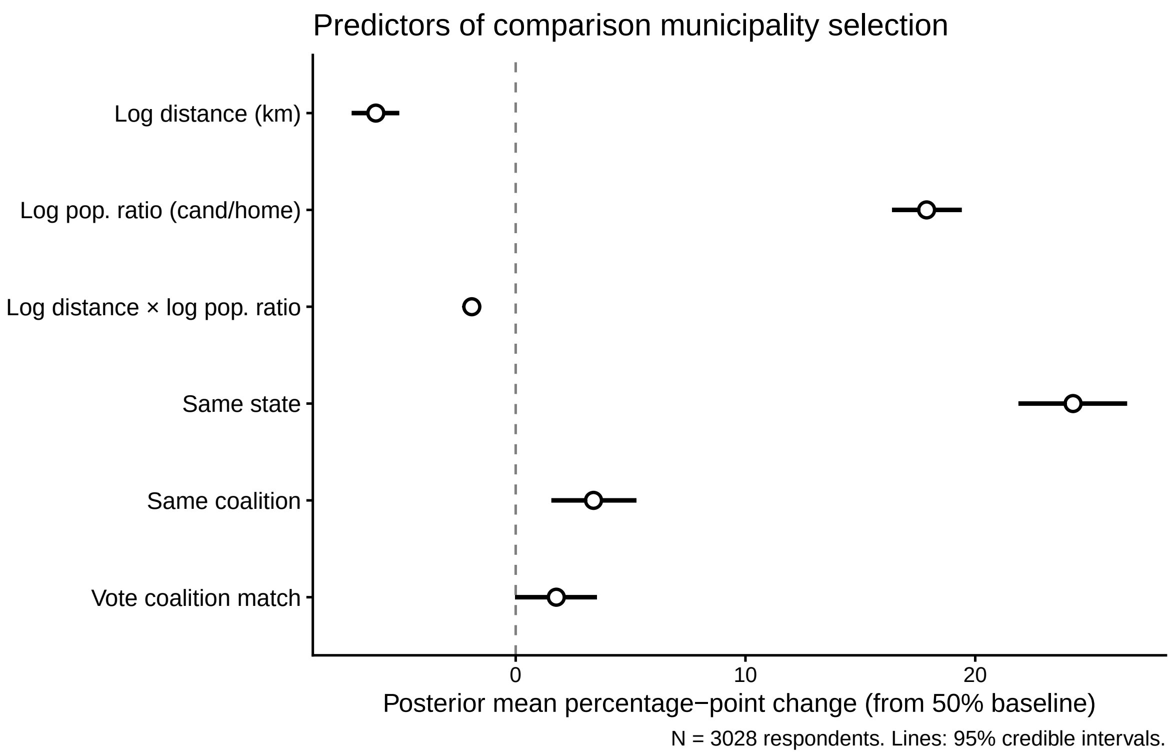


Figure 2: Analysis of the comparison municipality selections. Points are posterior means. Lines are 95% credible intervals. The reference category for coalition is MC. Pool variables and home municipality controls are not shown.

attenuated by distance. Third, respondents are more likely to select municipalities governed by the same coalition as their home municipality, suggesting that municipalities most relevant to respondents tend to share partisan government characteristics. Fourth, respondents are slightly more likely to select municipalities governed by the coalition that they report intending to vote for.

The preference for same-coalition municipalities is at least partially due to governing-party knowledge. I test this by examining respondents' knowledge of the governing coalition of comparison municipalities in wave 2, where some respondents were by chance assigned a comparison municipality that they chose in the first wave. Using linear probability models with respondent fixed effects, I find that respondents were 8 percentage points more accurate at identifying municipalities they selected as benchmarks in wave 1. These results suggest that knowledge of party governance did play a role in whether municipalities were selected by respondents (see Appendix 8 for full results).

Table 2: Bayesian multilevel logistic regression: predictors of benchmark municipality selection. Estimates are posterior means; Est. SD is the posterior standard deviation. Reference categories: MC (cand. and home coalition), random (pool). $N = 45{,}420$ observations, 3,028 respondents.

Parameter	**Estimate**	**Est. SD**	**l-95% CI**	**u-95% CI**	$\hat{R}$
Regression coefficients					
Intercept	−7.58	0.31	−8.20	−6.98	1.00
Log distance (km)	−0.24	0.02	−0.29	−0.20	1.00
Log pop. ratio (cand./home)	0.75	0.04	0.68	0.82	1.00
Same state	1.06	0.06	0.94	1.19	1.00
Same coalition (governing)	0.14	0.04	0.06	0.21	1.00
Vote–coalition match	0.07	0.04	−0.00	0.14	1.00
Cand. coalition: MORENA/PVEM/PT	0.02	0.06	−0.10	0.14	1.00
Cand. coalition: PAN/PRI/PRD	0.29	0.06	0.17	0.41	1.00
Cand. coalition: Other	−0.06	0.14	−0.35	0.21	1.00
Home coalition: MORENA/PVEM/PT	0.23	0.07	0.09	0.36	1.00
Home coalition: PAN/PRI/PRD	0.36	0.07	0.22	0.49	1.00
Log home pop.	0.39	0.02	0.35	0.43	1.00
Pool: nearest	1.38	0.07	1.23	1.52	1.00
Pool: largest	0.54	0.08	0.39	0.69	1.00
Log distance × log pop. ratio	−0.08	0.01	−0.09	−0.06	1.00
Random effects					
σ_α (respondent)	0.41	0.03	0.35	0.47	1.00

Experimental Design

In the experimental portion of this project, I test four treatment conditions that vary how crime information for a respondent's municipality is benchmarked. While I cannot manipulate the information that voters receive in real life, I can manipulate the information they receive in a survey experiment and determine whether it is "news" to them by comparing it to their pre-treatment beliefs.

Designing a survey experiment that matches real electoral conditions as closely as possible, however, is important for the external validity of the results (Incerti 2020; Breitenstein 2019). Thus, a balance must be struck between controlling for confounders and maintaining external validity. This paper attempts to strike this balance by using real performance information and eliciting voters' vote intentions and beliefs about politicians both pre- and post-treatment, which allows for testing of belief updating in a way that is more directly comparable to real-world conditions.

I present information on crime (specifically, robbery counts[3]) in the treatments. There are several reasons for this choice. First, crime is consistently among the top concerns of Mexican voters,[4] making it a high-salience issue. This is important, because a necessary condition for information's effect on voting is that it be relevant to voters (Bhandari, Larreguy, and Marshall 2023; Adida et al. 2020).

Second, municipal governments have genuine, if partial, responsibility for public security: municipal presidents control the funding, hiring, and structure of municipal police forces, which comprise the majority of law enforcement personnel in Mexico (Sabet 2012; Marshall 2024). Indeed, policy interventions have been shown to lower rates of robbery (Cabrera-Hernández and Zárate-Tenorio 2026). Even in cases where states have authority over municipalities (*mando único*), municipal presidents still generally control finances for public security (Víctor Cubillas 2025).[5] This makes crime a domain where voters can reasonably (partially) attribute crime outcomes to the incumbent.

Third, and most importantly for testing the theory, crime statistics are inherently noisy signals, a setting in which comparisons that filter noise are most useful. The data source is the Secretariado Ejecutivo del Sistema Nacional de Seguridad Pública (SESNSP), which reports monthly robbery counts at the municipal level. I use 2025 totals to maximize proximity to the 2027 municipal elections.

The first treatment arm (T1) provides plain information: a robbery rate per 100,000

3. I choose robberies instead of violent crime (like homicides) because the latter may be more associated with organized crime which is much harder to combat at the municipal level.

4. In a recent public opinion poll in El Economista (2025), crime was the top issue at 45.9%, while the next most popular issue (economy) was 9.0%

5. Only 3.5% of municipalities in 2022 spent 0 pesos on local public security according to the Censo Nacional de Gobiernos Municipales y Demarcaciones Territoriales de la Ciudad de México (CNGMD), although about 15% had missing data.

people for the respondent's own municipality, without any point of comparison. The second (T2) adds a bar chart comparing the respondent's municipality to a sample of similar municipalities, providing a non-partisan comparison. The third (T3) replaces that sample with municipalities governed by a political coalition (MORENA/PT/PVEM vs. PAN/PRI/PRD vs. MC) different than the coalition governing the respondent's municipality. The fourth (T4) uses municipalities governed by the same coalition as the respondent's municipality. The municipalities in the comparison arms (T2, T3, and T4) are selected using Mahalanobis matching on log population, log geographic area, and log distance from the respondent's home municipality. This matching strategy is implemented in order to show municipalities that are most relevant for comparison, and is consistent with the results from the descriptive analysis in Wave 1.

All treatment arms (T1–T4) present the following information: "Municipal presidents control police funding and, therefore, have partial influence over local crime. In [*respondent's municipality*], there were X robberies per 100,000 inhabitants in 2025." Comparison arms (T2–T4) add: "The following graph shows the robbery rate per 100,000 people in 2025 for [*respondent's municipality*] and a sample of similar municipalities..." In T2 (non-partisan comarison), no qualifier is added to the last sentcence. In T3 (opposition-coalition comparison), "...that are governed by [*opposite coalition*]" is added. In T4 (same-coalition comparison), "...that are governed by [*same coalition*]" is added. An example of the bar graph from a partisan comparison treatment group is shown in Figure 3. A bar graph from T2 would look identical but would not label which party the comparison municipalities are from.

The main control group (Control) provides a weather placebo: factual text about rainfall variation across Mexican municipalities, followed by the respondent's home municipality's average annual precipitation with no crime content or comparison. Specifically, the text of the control group reads: "Geography, altitude, and regional climatic systems are the primary determinants of precipitation patterns—factors that, to a large extent, we cannot

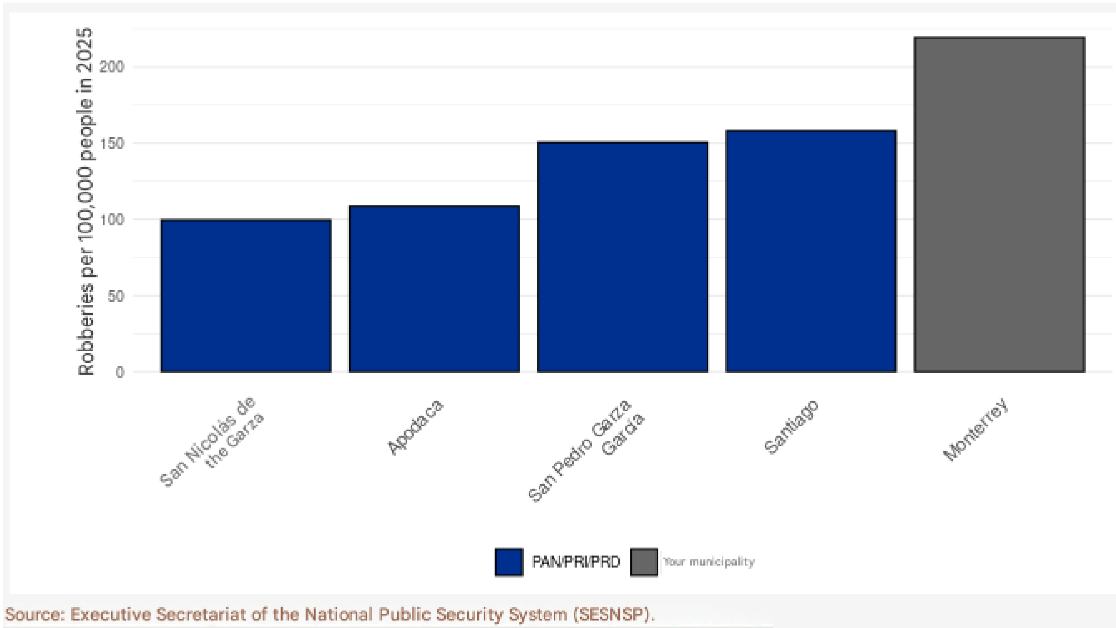


Figure 3: Example of a partisan comparison treatment graph, translated to English

control. In [respondent's municipality], there were X mm of precipitation in 2025." Control and comparison treatment arms were assigned with equal probability (0.2), while the plain information treatment arm was assigned a lower probability (0.1). An alternative control arm (Control2, assigned with probability 0.1) is used for robustness checks reported in the appendix. These arms were assigned lower probability because the main focus of the paper is on differences between benchmarks, relative to no-information controls. See the appendix for more details.

The arms are structured to address different aspects of how information is benchmarked. Specifically, the experiment in general tests whether information moves beliefs at all. The desigin also addresses whether the inclusion of a benchmark matters, and if it does, do differing benchmark designs have heterogeneous effects on voting behavior? These broad questions stem from the 8 pre-registered research questions in the pre-analysis plan.

Experimental Estimation Strategy

In estimating the effect of the informational treatments, heterogeneous treatment effects are of interest. Specifically, the estimation needs to account for whether the information is "good news" or "bad news" compared to baseline beliefs. To capture this, I define two complementary perception gap measures (Haaland, Roth, and Wohlfart 2023). Throughout this section i indexes respondents. The subscripts h and b from the theory correspond to respondent i's home municipality and the benchmark municipality shown, respectively.

The crime-level perception gap (CG_i) is defined as

$$CG_i = C_i^{\text{actual}} - C_i^{\text{prior}} \tag{2}$$

where C_i^{actual} is the respondent's home municipality's actual robbery rate per 100,000 inhabitants and C_i^{prior} is the respondent's pre-treatment estimate of robberies per 100,000 inhabitants in their municipality, elicited on the same scale. When eliciting the prior belief about the crime rate, respondents were informed of what the median municipality's crime rate was. When $CG_i > 0$, actual crime is higher than expected (bad news about the level of crime). In estimation, CG_i enters as a signed-log (log-modulus) transformation of this gap, $\text{sign}(CG_i) \times \ln(1 + |CG_i|)$. Respondents' priors have a long right tail, so untransformed gaps would let a small number of extreme values dominate the interaction terms. The crime-ranking perception gap (RG_i) is defined as

$$RG_i = R_i^{\text{actual}} - R_i^{\text{prior}} \tag{3}$$

where R_i^{actual} is the respondent's home municipality's robbery rank among comparison municipalities (higher rank = more robberies = worse relative performance) and R_i^{prior} is the respondent's pre-treatment perceived rank derived from the relative robbery ranking question. When $RG_i > 0$, the municipality's actual comparative standing is worse than the respondent expected.

The estimating equation is

$$y_i = \alpha_0 + \alpha_1 CG_i + \alpha_2 RG_i + \sum_{k=0}^{4} \delta_k T_{ik} + \sum_{k=0}^{4} \beta_k T_{ik} \times CG_i + \sum_{k=0}^{4} \gamma_k T_{ik} \times RG_i + \mathbf{X}_i' \boldsymbol{\lambda} + \varepsilon_i \quad (4)$$

where y_i is the post-treatment value of the outcome of interest for respondent i. $T_{ik} = 1$ if respondent i is assigned to arm k; arms are indexed $k \in \{1, 2, 3, 4\}$ corresponding to T1 (plain information), T2 (non-partisan benchmark), T3 (opposition benchmark), and T4 (same-coalition benchmark) respectively, with the non-comparison Control as the omitted baseline. $\mathbf{X}_i$ is a control for the prior, or the outcome measured pre-treatment. The key coefficients of interest are β_k, the treatment effect conditional on the crime level perception gap (good news or bad news about the level of crime), and γ_k, the treatment effect conditional on the crime ranking perception gap.

The two gap measures are not competing versions of the same quantity. Rather, they stand in for different objects in the model. CG_i captures surprise about the respondent's municipality's crime rate r_h. What varies across treatment arms is how much of that raw news the respondent can assign to the incumbent: in the same-coalition benchmark, both the common shock and the policy term difference out, so a given surprise carries more information about z_h than the same surprise does in the control, plain information, or non-partisan benchmark arms. The competence content of a treatment is thus carried by the interaction β_k rather than by CG_i on its own, which is why the theory makes predictions about β_k and not about α_1.

RG_i captures surprise about relative standing. Because the pre-treatment ranking question is asked about the same comparison municipalities the respondent is later shown, RG_i nets out what the respondent already expected the comparison to reveal. This measure is of particular interest in the opposition benchmark treatment arm, because according to the theory the opposition benchmark may rely on priors about how parties differ directionally.

Outcomes estimated in the model are vote intentions and beliefs about crime handling of incumbent governments and the three major coalitions. Specifically, I use intention to vote for the incumbent post-treatment as the main outcome and belief updating for explanation of the mechanism.

I verify that respondents are similar across treatme arms using equivalence tests (Hartman and Hidalgo 2018), which take substantial difference rather than equality as the null hypothesis and therefore provide positive evidence of balance rather than a mere failure to reject it. The arms are equivalent on every pre-treatment covariate except region, which fails only marginally and because of limited power in small regional cells rather than any observed imbalance (Appendix 5).

Experimental Results

Vote Intention Results

I first consider the effect of information on vote intention. Vote intention is a binary variable indicating whether a respondent reported intending to vote for the current incumbent coalition (1) or not (0). In models estimating this as the outcome, I control for respondents' pre-treatment vote intention. Figure 4 presents the experimental result with the vote intention outcome. The key finding is that only same-coalition benchmarks (T4) significantly change vote intention: respondents assigned to T4 who learn their municipality has higher crime than they expected are less likely to indicate they will vote for the incumbent party in the next local elections. This effect operates only through the crime-level perception gap (left panel), where T4 shows a statistically significant negative coefficient. All other treatment arms have null effects for both perception gaps.

Substantively, because the perception gap enters the model as a signed log, the size of the effect depends on the scale of the misperception rather than its raw magnitude. A one-standard-deviation increase in the logged perception gap is associated with a decrease of

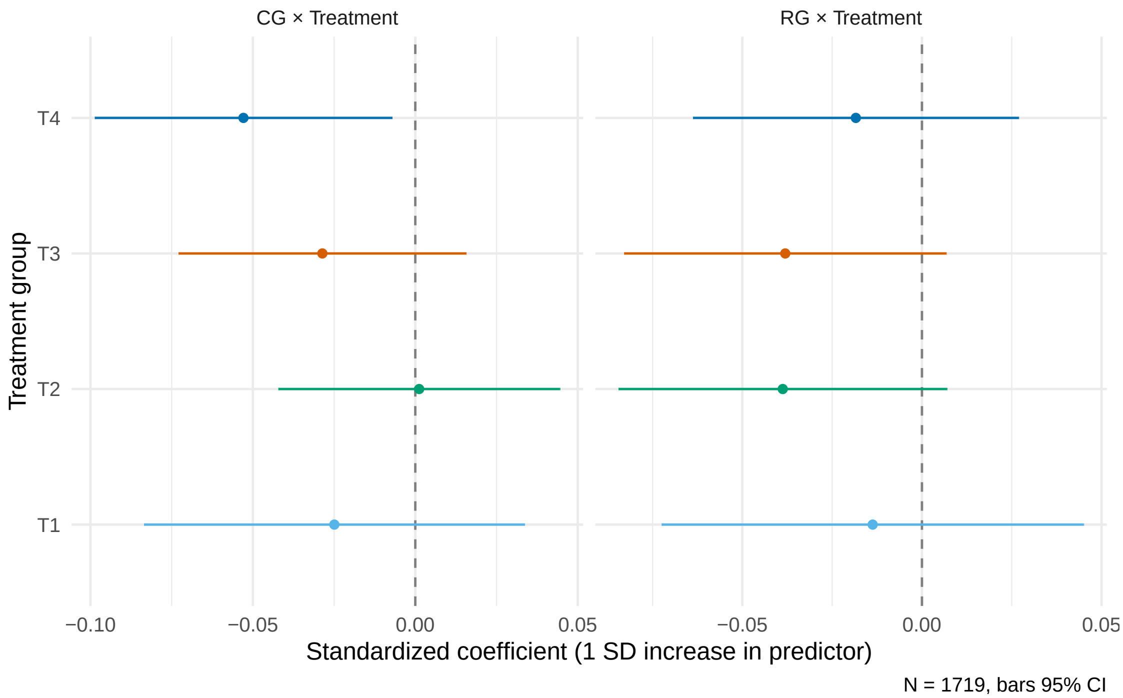


Figure 4: Coefficient estimates for the effect of each treatment on the change in vote intention for the respondent's home coalition. The left panel (CG $\times$ Treatment) plots the interaction of each treatment with the crime-level perception gap (β_k); the right panel (RG $\times$ Treatment) plots the interaction with the ranking perception gap (γ_k). Coefficients are standardized to a 1 SD increase in the respective gap measure. Points are OLS estimates; bars are 95% confidence intervals. The reference category is Control (weather placebo with no comparison).

approximately 5.3 percentage points in the probability of voting for the incumbent. The median respondent has a raw perception gap of 445, which on the logged scale sits roughly one standard deviation from accurate, so effects of this size describe a typical voter rather than an extreme one. Expressed differently, each doubling of the perception gap is associated with a 0.6 percentage point decrease in the probability of voting for the incumbent.

These results point to the case presented in the theory in which voters are more interested in learning about competence than policy differences, with one caveat: the theory predicts that the opposite-coalition treatment arm does reveals information about competence depending on voters' priors about how the two parties differ, whereas here the interaction that accounts for respondents' priors about relative performance ($RG \times T3$) shows no statistically significant effect. Additional evidence suggests this null results is due to noise in the full sample. The treatment does not tell respondents about which party governs their own municipality; this was done in order to not conflate information about own party governance with differences in benchmarks. When I subset the sample to only people who know which party governed their own municipality (roughly 80% of the sample), the RG interaction becomes statistically significant for the T3 treatment arm while the T4 general crime gap effect increases slightly in magnitude and maintains its statistical significance (see Appendix). Thus, restricting the sample to respondents for whom the partisan treatment is most effective strengthens the evidence in favor of an interpretation that voters are interested in voting out incompetent politicians.

The estimate from the same-coalition benchmark is not an artifact of respondents being especially familiar with the comparison municipalities they were shown. If the effect were driven by familiarity, the non-partisan benchmark rather than the same-coalition benchmark should show the larger effect, because respondents that were shown the non-partisan benchmark saw comparisons they found more familiar on both available measures. The municipalities shown in the non-partisan arm are closer to the benchmarks respondents chose for themselves in the descriptive analysis than those shown in the same-coalition arm. Respondents shown the non-partisan benchmark were also more likely to be shown a municipality they had explicitly selected in wave 1 (51% in T2 vs. 40% in T4). Appendix 7 provides more evidence that the effect is not driven by benchmark municipality familiarity alone.

Mechanism: Belief Updating

Given that the voting results imply respondents are focused on learning about incumbent competence, I now look at whether beliefs about how well politicians and parties handle crime explain the differences in voting intentions across treatment arms. Generally, if information causes voters to update negatively about how well an incumbent handles crime, they should be less likely to vote for the incumbent. More specifically, if the information forces voters to believe that the incumbent is worse than a challenger, they should be even less likely to continue to vote for the incumbent.

To test this, I examine the results of the model using beliefs about politicians as the outcome. Specifically, I estimate T4 (same-coalition comparison) interaction coefficients for two belief-update outcomes: evaluations on a 0–100 scale of the incumbent government's handling of crime ("Incumbent update"), and the incumbent's rating minus the average rating of the other coalitions ("Incumbent vs. other coalitions"). Both models are estimated on a common sample in a common form, so the estimates are directly comparable.

Figure 5 shows the coefficients from these models. Both point estimates are negative: respondents assigned to T4 who learn that crime is worse than they expected rate the incumbent, and the incumbent relative to the other coalitions less favorably. Only the relative outcome, however, is distinguishable from zero. A 1 SD increase in the crime-level perception gap lowers the incumbent's rating relative to the average of the other coalitions by roughly three points on the -100 to 100 scale (since it is the difference of two ratings on 0-100 scales), with a 95% confidence interval that excludes zero; the estimates for the incumbent alone is of comparable magnitude but imprecise enough that it is not distinguishable from zero.

Because the confidence intervals across outcomes overlap substantially, the difference in statistical significance is better read as a difference in precision than as firm evidence that the relative evaluation responds more strongly to the treatment. Still, the pattern is consistent with the vote-intention result. When crime information is benchmarked by same-coalition municipalities, respondents are more likely to negatively update beliefs about the incumbent government.

Table 3 reports the crime-level gap interaction for each treatment arm in the incumbent-versus-other-coalitions model. The table shows that T4's interaction is the largest in magnitude and the only coefficient distinguishable from zero. The difference between T4 and T3 is significant ($p = 0.01$), and the difference between T4 and T2 is borderline significant

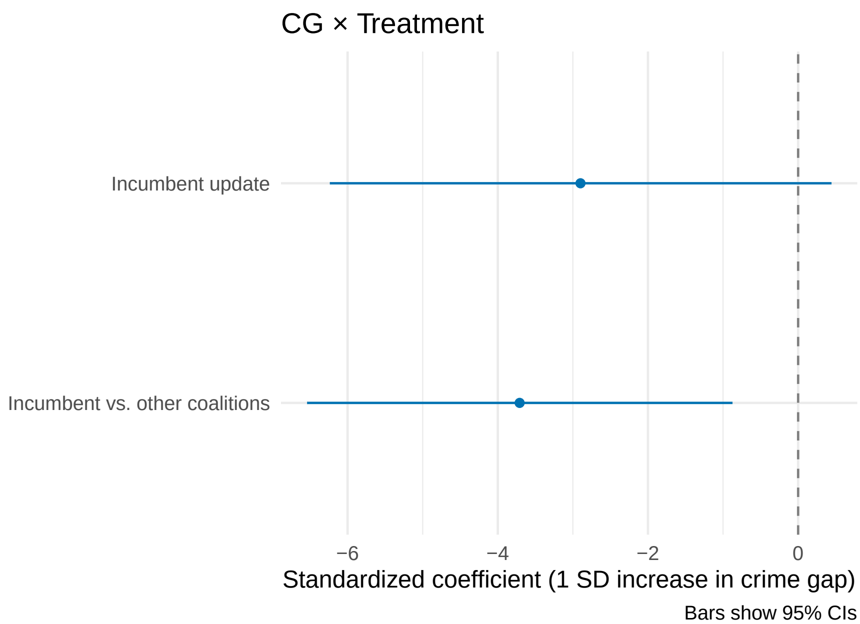


Figure 5: T4 (same-coalition comparison) $\times$ crime-level perception gap interaction coefficients (β_4) for three belief-update outcomes, each measured on a 0–100 slider: evaluations of the incumbent's handling of crime ("Incumbent update"), of the incumbent's party ("Home-party update"), and the incumbent's rating minus the average rating of the other coalitions ("Incumbent vs. other coalitions"). All three models are estimated on a common sample in a common form: the post-treatment outcome is regressed on the corresponding pre-treatment level(s) and the full set of gap $\times$ treatment interactions, so the three estimates are directly comparable. Coefficients are standardized to a 1 SD increase in the crime-level gap. Points are OLS estimates; bars are 95% confidence intervals. The reference category is Control (weather placebo with no comparison).

($p = 0.07$). This indicates that respondents assigned to the same-coalition treatment update more about the relative position of their incumbent government than respondents in other treatment arms.

Table 3: Incumbent rating minus the mean rating of the other coalitions (post-treatment levels), regressed on both pre-treatment levels, the perception gaps interacted with treatment arm, and pre-treatment coalition preference; only the crime-level gap (CG) $\times$ arm interactions are shown. Coefficients are standardized to a 1 SD increase in CG. HC2 robust standard errors.

Term	**Estimate**	**Std. Error**	p
CG $\times$ T1	-2.12	1.95	0.276
CG $\times$ T2	-0.77	1.46	0.599
CG $\times$ T3	0.67	1.52	0.658
CG $\times$ T4	-3.71	1.44	0.010
N	1704		
R^2	0.34		

Mediation Analysis

The results so far establish two things separately: same-coalition comparisons move vote intention, and they move beliefs about the incumbent relative to the other coalitions. They do not establish that the first happens *because of* the second. A treatment can shift a belief and shift a vote without the belief being what carries the vote. Mediation analysis splits the total effect of the same-coalition benchmark on incumbent vote intention into two pieces: the causal mediation effect and the direct effect. The causal mediation effect is the portion of the main effect that operates through a specified intermediate variable, and the direct effect is the portion of the effect that reaches the vote by other routes. Comparing their magnitudes says how much of the treatment's work the proposed mechanism is actually doing. This analysis was pre-registered as exploratory, and I note that the design lacks power to detect small mediation effects.

The mediator I use is the incumbent-versus-other-coalitions rating, the belief measure that most precisely responded to T4 in Figure 5 and thus the one for which a mediating role is

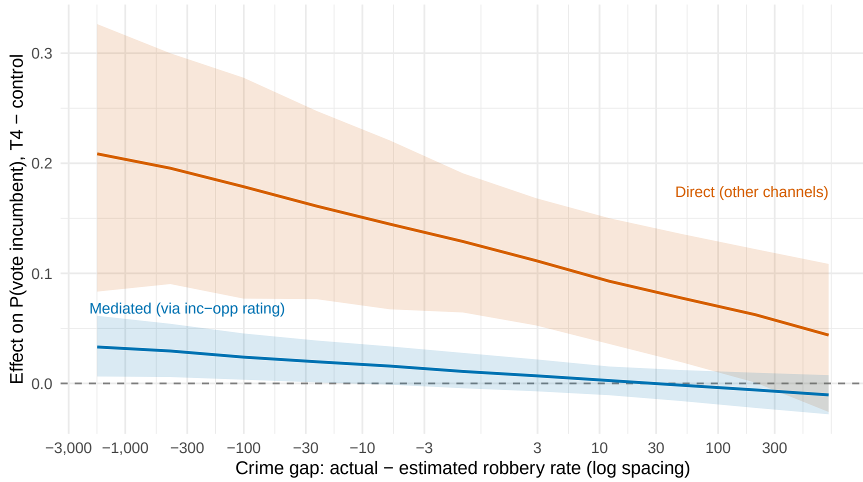


Figure 6: The T4 (same-coalition comparison) effect on the probability of voting for the incumbent coalition, relative to Control, decomposed into the portion transmitted through the incumbent-versus-other-coalitions rating ("Mediated") and the portion running through every other channel ("Direct"). Both are estimated across the range of the crime-level perception gap, with other variables held at their means. The horizontal axis is the signed log of the gap between the actual and the estimated robbery rate, labelled at nominal values. Estimates are quasi-Bayesian approximations (1,000 simulations) from pooled linear mediator and outcome models, each controlling for both pre-treatment ratings, the perception gaps interacted with arm, and pre-treatment coalition preference; bands are 95% confidence intervals.

most plausible. I estimate the decomposition using the two-equation approach of Imai, Keele, and Yamamoto (2010): one model for the mediator and one for vote intention, with treatment assignment, both perception gaps and their interactions with arm, and pre-treatment controls entering each. Because the treatments deliver good or bad news depending on what a respondent believed beforehand, a single average over respondents would obscure the pattern by averaging a positive and a negative effect. I therefore trace both quantities across the range of the crime-level perception gap, holding the ranking gap at its mean, with confidence intervals from a quasi-Bayesian approximation. Appendix 13 gives the full specifications, the potential-outcomes definitions of the ACME and ADE, and the assumptions under which they are identified.

Figure 6 decomposes the same-coalition benchmark effect on incumbent vote intention into the part that travels through the incumbent-versus-other-coalitions rating and the part that does not. Both components slope downward as the news worsens, as expected: information raises incumbent support when respondents learn crime is better than they had assumed and erodes it when they learn the opposite. At the good-news end of the plotted range (a gap of roughly 2,650 robberies fewer than the respondent expected), the mediated component is 0.03 with a confidence interval of [0.00, 0.06]. At the bad-news end the mediated component falls to -0.01, and is not statistically distinguishable from zero with a 95% confidence interval of [-0.03, 0.01]. The direct component is larger at all points in the range.

The results provide suggestive evidence that same-coalition benchmarks do move beliefs about the incumbent relative to the alternatives, and those beliefs do carry some of the effect on vote intention in the expected direction. However, this particular mediation pathway is only a small part. Respondents may also be updating about the incumbent's general competence rather than on crime specifically. Additionally, note that the mediated component's confidence interval covers zero across nearly the whole range, so its magnitude is better described as small and imprecisely estimated than as reliably positive.

Discussion

In this paper, I have shown that the way information is benchmarked can have a meaningful effect on how voters update their beliefs and intentions. Same-coalition benchmarks, in particular, provide a credible point of comparison that allows voters to attribute outcomes to incumbent performance rather than partisan differences. This has implications for both the design of informational interventions and our understanding of voter behavior in contexts where information is noisy and partisan cues are salient.

When designing informational campaigns, policymakers and researchers should consider not just the content of the information but also how it is benchmarked. Providing voters with relevant benchmarks that align with their political context can enhance the effectiveness of information dissemination and potentially influence electoral outcomes. How to determine which benchmarks are most relevant may differ from context to context, so researchers should not assume that the results in this paper will apply to contexts with different political contexts. Prior to fielding informational treatments, researchers should explore empirically determine which factors that are relevant to voters in order to maximize the credibility and salience of benchmarked information.

More broadly, these findings contribute to the literature on political information and voter behavior by highlighting the importance of credible comparisons for making information interpretable and indicative of government performance. In contexts where voters have limited information or face noisy signals, providing clear and relevant benchmarks can help them make more informed choices. Whether media coverage, political campaigns, or civic education programs present information in a way that allows voters to make meaningful comparisons may be just as important as the information itself.

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